JVET 35차 표준화 회의 출장 보고서

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1. 출장 개요

1.1. 출장자

- 소속: 동아대학교 컴퓨터 AI 공학부 영상신호처리연구실
- 성명: 김선재/이태식 박사과정, 박병주/이동호 석사과정

1.2. 출장지 및 기관

– 일본 삿포로 / ACU Convention Center, Sapporo Convention Center

1.3. 출장 목적

– JVET 35 차, MPEG 147 차 국제 표준화 회의 참석

1.4. 출장 기간

– 총 출장 기간
  - 김선재/이태식 박사과정 (8 박 9 일): 2024 년 7 월 11 일 (목) ~ 7 월 19 일 (금)
  - 박병주/이동호 석사과정 (9 박 10 일): 2024 년 7 월 11 일 (목) ~ 7 월 20 일 (토)

– 표준화 회의 기간 (7 박 8 일): 2024 년 7 월 12 일 (금) ~ 7 월 19 일 (금)

1.5. 관련 과제 및 수행 업무

생략 (공개용 보고서)

1.6. 수행과제와의 연계성(출장목적)

생략 (공개용 보고서)

1.7. 세부 일정 및 활동 내용

일 시	일정 계획 및 회의 안건
2024.07.11 (목)	- 김선재/이태식 박사과정: 출국 • 김해공항 → 삿포로 치토세 공항: 13:40 ~ 16:10 (2 시간 30 분) - 박병주/이동호 석사과정: 출국 • 김해공항 → 삿포로 치토세 공항: 09:05 ~ 11:40 (2 시간 35 분)
2024.07.12 (목) ~ 2024.07.19 (금)	Fri. 12 July, 1st day • 0910–1010 Opening remarks, review of practices, agenda, IPR policy reminder • 1010–1300 Reports of AHGs 1–14 • 1430–1500 AHG15/AHG16 reports • 1500–2100 EE1 and EE2 summary reports, EE2-1.x…EE2-3.x Sat. 13 July, 2nd day • 0900–1300 EE1/NNVC • 1430–1945 EE2 summary report, EE2-4.x…EE2-6.x Sun. 14 July, 3rd day • 0900–1300 EE2 related / non-EE2 ECM ➢ **기고 발표 진행 (발표자: 이태식, 기고서: JVET-AI0076)** • 1430–2055 EE2/ECM (Y. Ye from 1800) Mon. 15 July, 4th day • 0900–1200 MPEG WGs information sharing (outside of JVET) • 1440–1515 JVET plenary: Coordination, planning • 1515–1830 EE2/ECM (JRO/Y. Ye)

	Tue. 16 July, 5th day • 0900–1100 BoG on NNVC SW and EE1 planning (E. Alshina, F. Galpin) • 1430–1530 Joint meeting with WG 2 and VCEG on future video standardization (2 documents in 4.1) • 1545–1700 remaining 4.x except 4.5-4.7 • 1800–2000 EE2 remeining 5.2.4, 5.2.5 (Yan Ye) Wed. 17 July, 6th day • 0900–1000 MPEG information sharing session (outside of JVET) • 1015–1130 Joint meetings with AG 5 on visual tests and other issues (4.5-4.6) • 1130–1200 with WG 4 on VCM CTC • 1200–1300 4.16 GFVC • 1430–1730 NNVC BoG, remaining 4.x (except 4.1, 4.5-4.6, 4.11, 4.14) Thu. 18 July, 7th day • 0900–1345 Remaining doc review, revisits, output and AHG planning, etc. • 1500–1800 Output and AHG planning, BoG report gaming, remaining doc review, revisits, etc. Fri. 19 July, 8th day • 0920–1235 Wrap up plenary: • Remaining revisit(s) • Establishment of AHGs • Review and approval of output docs • Software timelines • Review of WG 5 meeting recommendations • Future planning, a.o.b.
2024.07.19 (금)	- 김선재/이태식 박사과정: 귀국 • 삿포로 치토세 공항 → 김해공항: 17:10 ~ 19:40 (2 시간 30 분)
2024.07.20 (토)	- 박병주/이동호 석사과정: 귀국 • 삿포로 치토세 공항 → 김해공항: 12:40 ~ 15:30 (2 시간 50 분)

2. 주요 활동 내용 (JVET)

2.1. AHG11 Neural Network-based Video Coding (NNVC)

2.1.1. 회의 내용 및 결과

❍ **AHG report 검토 [JVET-AI0011] [Co-chairs: E. Alshina (Huawei), F. Galpin (InterDigital), S. Liu (Tencent), A. Segall (Amazon), Vice-chairs: J. Li (ByteDance), R.-L. Liao (Alibaba), D. Rusanovskyy (Qualcomm), M. Santamaria (Nokia), T. Shao (Dolby), M. Wien (RWTH), P. Wu (ZTE)]**

- Evaluate and quantify the performance improvement potential of NN-based video coding technologies compared to existing video coding standards such as VVC, including both individual coding tools, architectures and content adaptation with NN

parameters overfitting.

- Discuss potential refinements of the test conditions for NN-based video coding in JVET-AF2016. Generate and distribute anchor encoding, and develop supporting software as needed.
- Study the impact of training (including the impact of loss functions) on the performance of candidate technologies and identify suitable material for testing and training.
- Analyze complexity characteristics for technologies under study, including transformers, perform complexity analysis, and develop complexity reductions of candidate technology.
- Finalize and discuss the EE on neural network-based video coding.
- Promote the call for training materials, distribute it, and actively communicate with content owners.
- Coordinate with other groups, including SC 29/AG 5 on the evaluation and assessment of visual quality, and AHG12 on the interaction with ECM coding tools. If possible, prepare encodings with combinations of tools included in the NNVC software for visual quality assessment at the next meeting.
- Coordinate with AHG14 on items related to NNVC software development and studying the impact of training set extension.

❍ **NNVC Software Development AHG14 report 검토 [JVET-AI0014] [F. Galpin (InterDigitla), Y. Li (ByteDance), Y. Li (Qualcomm), J. N. Shingala (Ittiam), R. Chang (Tencent), Z. Xie (OPPO)]**

- NNVC software 신규 버전 (NNVC-9.0) 개발에 따라 NNVC 채택 기술들에 대한 성능 비교 수행
  - ➢ **NNVC9.0**
  1. 신규 채택 기술 및 변경내용: (LOP3): training scripts + models + NNVC changes, (LOP3): models, HOP3, (improved MCTF, port from ECM), (rpr 1.5), VLOP, LOP2 adaptive content: training scripts, new SADL
  2. Bug-fix 및 SW cleanup: Build for ARM, Fixes many implicit cast to pass MSVC strict build, Clean model name, Add support for Y4M, Fix memory issue with padding
  - ➢ **NNVC9. 1**
  1. 신규 채택 기술 및 변경내용: LOP2 adaptive content: models and NNVC inference, HOP4 models + training scripts + NNVC changes, Extend SPS model ID to support new models, SADL update for HOP4 support

2. Bug-fix 및 SW cleanup: Fixes on issues avec NNVC-9.0 integration, Fixes legacy memory issues from VTM, fix parameter value for training

- LOP + NN-intra, HOP + NN-intra에 대한 실험 결과 제시 (Anchor: NNVC-VTM version)
- Encoding, decoding time은 anchor 대비 시간을 배수로 표현하였으며, CPU 기준 시간을 계산함

Test	# Par M	kMAC /pxl	Random Access cfg.					All Intra cfg.				
			Y	U	V	Enc	Dec	Y	U	V	Enc	Dec
LOP	N/A	N/A	**-7.31 %**	-13.11%	-11.32%	1.22	81.13	**-8.39%**	-14.17%	-13.86%	1.72	48.62
HOP	N/A	N/A	**-14.06 %**	-19.16%	-19.55%	2.87	1446	**-13.31%**	-16.16%	-17.71%	2.57	N/A
VLOP	N/A	N/A	**-5.30 %**	-5.43%	-5.16%	1.18	29.91	**-7.05%**	-7.82%	-7.74%	1.60	25.19

❍ **EE (Exploration experiment) Activity 검토 [JVET-AI0023] [E. Alshina (Hauwei), R. Chang (Tencent), F. Galpin (InterDigital), Y. Li (ByteDance), D. Rusanovskyy (Qualcomm), M. Santamaria (Nokia), J. Ström (Erricson), Z. Xie (OPPO) (EE coordinators)]**

- 직전 회의 (JVET-AG)에서 아래와 같은 EE를 진행하기로 결정하였으며, 금번 회의에서 해당 EE 결과들을 보고함
  - ➢ EE1-1: LOP and VLOP in-loop filter
    1. EE1-1.1: LOP input adjustment with trainable components (JVET-AI0173)
    2. EE1-1.2: Partial Convolution and Over-Parameterization (JVET-AI0068)
    3. EE1-1.3: On Low Complexity Operational Point for In-Loop Filtering (JVET-AI0221)
    4. EE1-1.4: Content-adaptive VLOP-filter (JVET-AI0157)
    5. EE1-1.5: VLOP using LOP3 architecture (JVET-AI0107)
  - ➢ EE1-2: HOP in-loop filter
    1. EE1-2.1: On the detailed analysis of integer Transformer ILF with luma-chroma component balancing and complexity trade-off (JVET-AI0176)
    2. HOP4+: On the complexity adjustment of HOP4 (JVET-AI0172)
  - ➢ EE1-3: NN-inter prediction
    1. withdrawn
  - ➢ EE1-4: NN-super-resolution
    1. EE1-4.1: Multiple scaling ratios coding for NNVC with NNSR (JVET-AI0074)

2. EE1-4.2: On simplified super resolution model (JVET-AI0093)

➢ EE1-5: Neural network based Intra coding

1. EE1-5.1: Combination of the neural network-based intra prediction mode and ISP (JVET-AI0130)

- 각 EE들에 대한 결과 요약은 아래 표와 같음 (Anchor: NNVC VTM mode)

➢ All Intra cfg BD-rate (vs VTM)

Test	All Intra (Summary)					Total kMAC/pxl	Total Param (Mprm)
	Y	U	V	Enc	Dec		
NN-Intra & LOP NN-filter							
NNVC-9.1(LOP3)	-8.4%	-14.2%	-13.9%	1.7	49	24.8	1.72
NNVC-8.0(LOP2)	-8.1%	-13.3%	-13.4%	1.7	44	24.9	1.55
NNVC-7.1(LOP1)	-8.1%	-13.3%	-13.4%	1.8	56	24.9	1.55
NN-Intra & VLOP NN-filter							
NNVC-9.1(VLOP)	-7.1%	-7.8%	-7.7%	1.8	29	15.0	0.4
NN-Intra & LOP NN-filter & Adaptive resolution coding NNVC-8.0 (*)							
NNVC-9.1(HOP4)	-13.3%	-16.2%	-17.7%	2.6	853	483.6	3.0
NNVC-8.0(HOP3)	-12.7%	-11.6%	-13.0%	2.4	56	473.8	2.9
NN-Intra & HOP NN-filter							
NNVC-8.0 RPR(*)	-8.5%	-11.3%	-11.0%			24.9	1.55
NNVC-8.0 NNSR(*)	-8.6%	-11.8%	-11.3%			45.2	1.63
(*) encoder problem for adaptive resolution coding in NNVC-9.1							

Test	All Intra cfg BD-rate vs VTM					# tools combined	kMAC /pxl	Total Num Param
	Y	U	V	Enc	Dec			
NN-Intra & LOP NN-filter								
NNVC9.1-LOP	0.0%	0.0%	0.0%	1.7	49	2	24.8	1.72
EE1-1.1	0.0%	0.0%	0.5%			2	26.8	1.72
EE1-1.2.1	0.0%	2.3%	2.3%			2	24.8	1.5
EE1-1.2.2	0.0%	-1.2%	-1.7%			2	24.8	1.5
EE1-1.2.3	0.0%	-1.6%	-2.6%			2	24.8	1.5
EE1-1.2.3(int)	0.1%	-1.7%	-2.5%			2	24.8	1.5
EE1-1.3.a, fl32	-0.7%	1.4%	1.3%			2	23.5	1.5
* EE1-1.1 – trainable 'DCT' for LOP filter * EE1-1.2.1 – over parametrization (training hint) * EE1-1.2.2 – partial convolution (total number of convolutions 94 74) * EE1-1.2.3 – over parametrization and – partial convolution * EE1-1.3.a – new architecture for LOP filter (full float32 results made available on July 9th 2024)								
NN-Intra & VLOP NN-filter								
NNVC9.1-VLOP	0.0%	0.0%	0.0%	1.8	29	2	15.0	0.4
EE1-1.3b, fl32	-0.6%	-1.5%	-1.3%				12.6	1.5
EE1-1.5	-0.1%	-1.4%	-0.9%				13.1	1.5
* EE1-1.3.b – new architecture for VLOP filter (full float32 results made available on July 9th 2024). * EE1-1.5 – VLOP with LOP architecture								
NN-Intra & HOP NN-filter								
NNVC9.1-HOP	0.0%	0.0%	0.0%	2.6	853	2	483.6	3.0
EE1-2.1.1	-1.1%	3.4%	3.7%				476	2.89
EE1-2.1.4	-0.1%	-2.2%	-1.3%				473	2.85

HOP4+	-0.2%	0.3%	0.7%				474	1.5
* EE1-2.1.1 – transformers in all BBBs. * EE1-2.1.2 – transformers in two BBBs. 'HOP4+' design is recommended during AhG11&14 teleconference								
NN-Intra & LOP NN-filter & Adaptive resolution coding NNVC-8.0 (*)								
NNVC-9.1	-8.4%	-14.2%	-13.9%			3	24.9	1.55
NNVC-8.0 NNSR	-8.6%	-11.8%	-11.3%				45.2	1.79
NNVC-9.1-NNSR(*)	-8.6%	-11.8%	-11.3%			3	45.2	1.63
EE1-4.1.1 (RPR)	-8.8%	-11.2%	-10.9%				24.8	1.73
EE1-4.1.2 (NNSR)	-9.2%	-11.5%	-11.0%				29.5	1.78
EE1-4.2	-9.0%	-12.6%	-11.8%				30.1	1.77
* EE1-4.1.1 Encoder strategy for adaptive resolution (RPR filter for re-sampling) * EE1-4.1.2 same as EE1-4.1.1 but with NNSR filter (simpler than one in NNVC-9.1) * EE1-4.2 NNSR but simpler than one in NNVC-9.1. * BD-rate gain measured based on MS-SSIM for adaptive resolution coding is 2…2.2% higher compared to PSNR gain (both for RPR and NN-based filters) (*) encoder problem for adaptive resolution coding in NNVC-9.1								
NN-Intra								
NNVC9.1	0.0%	0.0%	0.0%				24.8	1.72
NNVC9.1-LCnn Intra	0.4%	0.3%	0.3%				21.7	1.53
EE1-5.1	-0.1%	-0.2%	-0.1%				21.7	1.53
* EE1-5.1 improved NN-INTRA with ISP design and better loss-function in training.								

➢ Random Access cfg BD-rate (vs VTM)

Test	Random Access (Summary)					Total kMAC/pxl	Total Param (Mprm)
	Y	U	V	Enc	Dec		
NN-Intra & LOP NN-filter							
NNVC-9.1(LOP3)	-7.3%	-13.1%	-11.3%	1.2	81	24.8	1.72
NNVC-8.0(LOP2)	-6.9%	-13.2%	-12.1%	1.2	73	24.9	1.55
NNVC-7.1(LOP1)	-6.9%	-13.2%	-12.1%	1.3	86	24.9	1.55
NN-Intra & VLOP NN-filter							
NNVC-9.1(VLOP)	-5.3%	-5.4%	-5.2%	1.2	40	15.0	0.4
NN-Intra & LOP NN-filter & Adaptive resolution coding NNVC-8.0 (*)							
NNVC-9.1(HOP4)	-14.0%	-19.1%	-19.5%	2.6	853	483.6	3.0
NNVC-8.0(HOP3)	-13.7%	-13.9%	-14.5%	2.4	56	473.8	2.9
NN-Intra & HOP NN-filter							
NNVC-8.0 RPR(*)	-7.5%	-10.9%	-9.7%			24.9	1.55
NNVC-8.0 NNSR(*)	-7.8%	-11.9%	-10.5%			45.2	1.63
(*) encoder problem for adaptive resolution coding in NNVC-9.1							

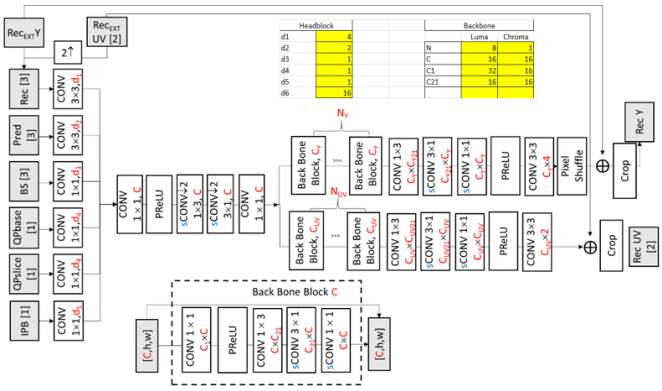
Test	All Intra cfg BD-rate vs VTM					# tools combined	kMAC /pxl	Total Num Param
	Y	U	V	Enc	Dec			
NN-Intra & LOP NN-filter								
NNVC9.1-LOP	0.0%	0.0%	0.0%	1.2	81	2	24.8	1.72
EE1-1.1	-0.1%	0.5%	0.7%				26.8	1.72
EE1-1.3.a, fl32	-0.8%	1.2%	0.1%				23.5	1.5
* EE1-1.1 – trainable 'DCT' for LOP filter * EE1-1.3.a – new architecture for LOP filter (full float32 results made available on July 9th 2024)								
NN-Intra & VLOP NN-filter								

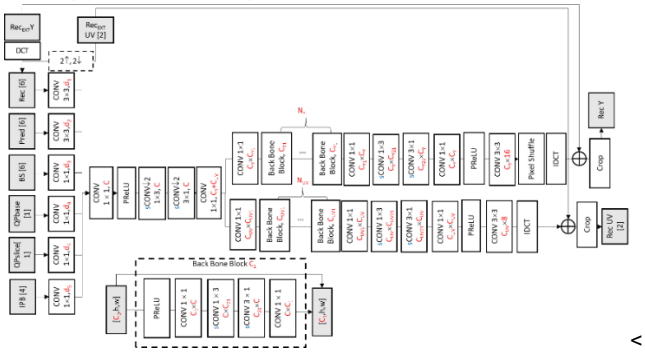
NNVC9.1-VLOP	0.0%	0.0%	0.0%	1.2	40	2	15.0	0.4
EE1-1.3b, fl32	-0.7%	-1.5%	-1.2%				12.6	1.5
EE1-1.4	-1.0%	-1.7%	-1.0%				13.4	1.5
EE1-1.5	-0.2%	-1.6%	-0.7%				13.1	1.5
* EE1-1.3.b – new architecture for VLOP filter (full float32 results made available on July 9th 2024). * EE1-1.4 – content adaptive VLOP. * EE1-1.5 – VLOP with LOP architecture								
NN-Intra & HOP NN-filter								
NNVC9.1-HOP	0.0%	0.0%	0.0%	2.9	1446	2	483.6	3.0
EE1-2.1.1	-1.2%	1.6%	0.7%				476	2.89
EE1-2.1.4	-0.1%	-2.9%	-3.3%				473	2.85
HOP4+	-0.1%	-0.2%	-0.4%				474	1.5
* EE1-2.1.1 – transformers in all BBBs. * EE1-2.1.2 – transformers in two BBBs. 'HOP4+' design is recommended during AhG11&14 teleconference								
NN-Intra & LOP NN-filter & Adaptive resolution coding NNVC-8.0 (*)								
NNVC-9.1	-7.3%	-13.1%	-11.3%			3	24.9	1.55
NNVC-8.0 NNSR	-7.8%	-12.8%	-10.5%	-8.6%			45.2	1.79
NNVC-9.1-NNSR(*)	-7.8%	-11.9%	-10.5%			3	45.2	1.63
EE1-4.1.1 (RPR)	-7.9%	-10.9%	-8.7%	-8.8%			24.8	1.73
EE1-4.1.2 (NNSR)	-8.1%	-10.3%	-8.9%	-9.2%			29.5	1.78
EE1-4.2	-8.1%	-11.4%	-9.4%	-9.0%			30.1	1.77
* EE1-4.1.1 Encoder strategy for adaptive resolution (RPR filter for re-sampling) * EE1-4.1.2 same as EE1-4.1.1 but with NNSR filter (simpler than one in NNVC-9.1) * EE1-4.2 NNSR but simpler than one in NNVC-9.1. * BD-rate gain measured based on MS-SSIM for adaptive resolution coding is 2…2.2% higher compared to PSNR gain (both for RPR and NN-based filters) (*) encoder problem for adaptive resolution coding in NNVC-9.1								
NN-Intra								
NNVC9.1	0.0%	0.0%	0.0%				24.8	1.72
NNVC9.1-LCnn Intra	0.1%	0.0%	0.0%				21.7	1.53
EE1-5.1	-0.1%	-0.5%	-0.4%				21.7	1.53
* EE1-5.1 improved NN-INTRA with ISP design and better loss-function in training.								

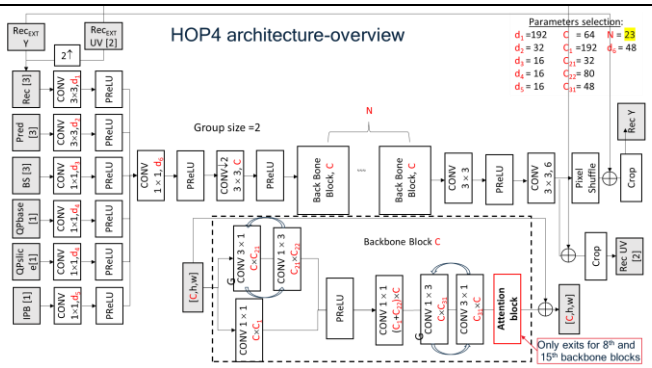
* Run time is not always reliable.

- 금번 회의에서 발표한 루프내 필터 관련 NNVC EE 결과의 주요 결정사항은 아래와 같음
  - VLOP EE1-1.4 Adopt: Content adaptive VLOP
  - VLOP EE1-1.5 Adopt: VLOP with LOP architecture
  - HOP HOP4+ Adopt: High-performance Operation Point (HOP)
  - NNSR EE1-4.1.2 Adopt: Multi scaling ratios coding NNSR
  - NN-Intra EE1-5.1 결과 확인 후 Adopt: Combination of the neural network based intra prediction mode and ISP

- 금번 회의에서 채택된 루프내 필터 기술들의 결과 및 요약은 아래 표와 같음

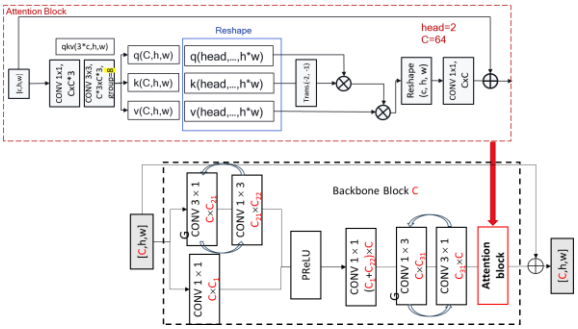
EE1	Title	Contributors	Reviews
1.4 (VLOP)	Content-adaptive VLOP-filter (JVET-AI0157)	R. Yang, M. Santamaria, F. Cricri, M.M. Hannuksela, D.Bugdayci Sansli, A.Hallapuro, J.Lainema, H. Zhang (Nokia), D.Rusanovskyy, Y.Li, M.Karczewicz (Qualcomm)	**Summary** • The content-adaptive VLOP filter is obtained by **overfitting multiplier parameters** that are applied after a convolutional block**.** •Formula: σ((W*x+b)·m), where W is the kernel, * is the convolution operator, x is the input, b is the bias, m is the multiplier and σ is the activation function. • **The network architecture is based on JVET-AH0051**, and the **difference is that the multiplier layers** are added after each convolutional block in the luma and chroma backbone. • Similar to the content-adaptive loop filter in JVET-AH0096, only subsets of multipliers in **the luma and chroma backbones are selectively overfitted.** • The number of updated multiplier parameters is defined through RDO, varies between coded bitstreams, and is defined adaptively at the overfitting stage. • The details of selection and determination of **the number of overfitted multiplier layers are described in JVET-AH0096.**  < VLOP filter architecture from JVET-AH0051 > **Recommendations** • It was pointed out that an implementation as post filter, and enabling by NNPF SEI would also be desirable.
1.5 (VLOP)	VLOP using LOP3 architecture (JVET-AI0107)	D. Liu, J. Ström, M. Damghanian, P. Wennersten (Ericsson), D. Rusanovskyy, Y. Li, M. Karczewicz (Qualcomm)	**Summary** • In this contribution, we use the **same architecture as LOP3** (JVET-AH0051) and modify the number of channels and backbones blocks to meet the VLOP complexity. • Block changes

			Head block: • d1: 4→8 • d2: 2→4 • d3: 1→2 • d4: 1 • d5: 1 • d6 (=C): 16→64 **Backbone block:** • Luma ○ N_Y: 8→6 ○ C_Y: 16→32 ○ $C1_Y$: 32→96 ○ $C21_Y$: 16→32 • Chroma ○ N_{UV}: 1 ○ C_{UV}: 16 ○ $C1_{UV}$: 16→32 ○ $C21_{UV}$: 16 •Note that transformed inputs are used, and before splitting into luma and chroma branches, the channel size is CY + CUV. • The complexity of the proposed VLOP model is 5.16 kMAC/pixel, and the number of parameters is 60686. • As a comparison, the current VLOP has a complexity of 5.18 kMAC/pixel and 16005 parameters. • **kMAC/pixel 0. increase, parameter 3.8x**  < The architecture of LOP3 >
HOP4+ (HOP)	On the complexity adjustment of HOP4 (JVET-AI0172)	Y. Li, D. Rusanovskyy, M. Karczewicz (Qualcomm), F. Galpin (Interdigital), Y. Li, J. Li, C. Lin, K. Zhang, L. Zhang (Bytedance)	**Summary** • A unified filter architecture for the High-performance Operation Point (HOP) was defined in JVET-AD0380. • Updated training strategy and configuration were presented in JVET-AG0041. • The HOP4 software integration, combining with Attention module (JVET-AH0205) and wide activation of JVET-AH0189 was jointly trained by the proponents and presented during AhG11/14 telco. • The proposed configuration is straightforward. Based on the HOP4 architecture, the 3x3 group convolution in the attention block is set to a group of 8, and the number of residue block is set to N=23.



< The architecture of HOP4 >

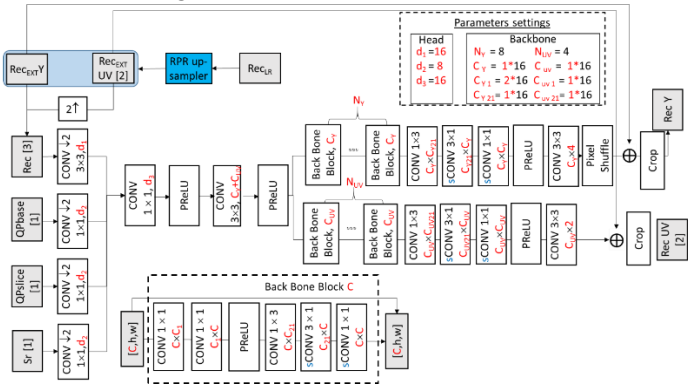
HOP4 architecture - Attention block

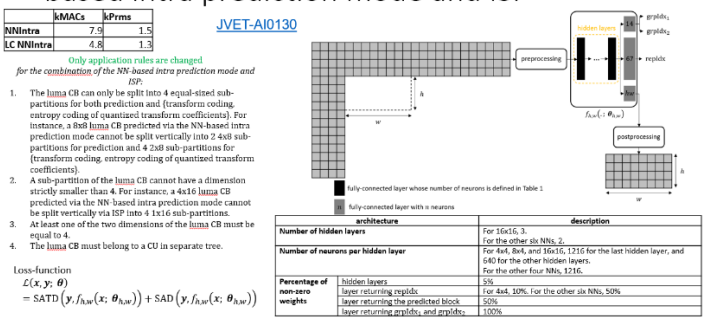


< The architecture of HOP4 attention block >

Recommendations

• HOP4+ is well understood, and provides benefit against the previous HOP designs. → next HOP design

4.1 (NNSR)	Multiple scaling ratios coding for NNVC with NNSR (JVET-AI0074)	Z. Lv, C. Zhou (vio)	**Summary** • NVC9.0은 i번째 GOP의 첫 번째 사진의 다운-업 스케일된 PSNR을 임계값과 비교하여 i번째 GOP를 2배 축소할지 적응적으로 결정할 수 있다. • However, the current NNVC does not have such functionality to support switching within multiple scaling ratios. • 이전 기여들과 비교하여, 세 가지 스케일 팩터 {x2.0, x1.5, x1.0}를 지원. • Scaling ratio (*Sr*) is set as an input to the model: - For scaling ratio is 2.0x, *Sr* = 1; - For scaling ratio is 1.5x, *Sr* = 2; - For scaling ratio is 1.0x, *Sr* = 0.  < Network structure > **Recommendations** • 4.1.2 and 4.2 have comparable performance and complexity (minor advantage of 4.1.2 could be seen), but 4.1.2 has additional benefit of arbitrary scaling factors.
5.1 (NN-Intra)	Combination of the neural network-based intra prediction mode and ISP (JVET-AI0130)	T. Dumas, F. Galpin, P. Bordes (Interdigital)	**Summary** • NN 기반 인트라 예측 모드와 ISP의 결합에서 ISP는 루마에만 적용되며, 수평 및 수직 분할 유형을 사용 • NN 기반 인트라 예측 모드로 예측된 루마 CB는 4개의 동일 크기 하위 파티션으로만 분할될 수 있으며, 하위 파티션은 크기가 4보다 작을 수 없고, 두 차원 중 하나는 반드시 4여야 하며, 별도 트리에 속해야 함 • 비 NN 인트라 예측 모드의 경우 ISP와 LFNST가 사용될 때 LFNST 암시적 신호는 인트라 예측 모드의 인덱스에 의해 결정되며, NN 기반 인트라 예측 모드의 경우 각 루마 TB에 대해 NN이 LFNST 암시적 신호를 추론 • 인코더 측에서 비 NN 인트라 예측 모드와 ISP의 결합이 테스트되면, 인덱스는 ISP 목록 내 동일한 위치에 배치되며, NN 기반 인트라 예측 모드와 ISP의 결합이 허용되면 인덱스가 첫 번째 위치에 배치 됨 • 인트라 예측을 위한 NN 훈련의 목적 함수는 원본 블록과 NN 예측 사이의 SATD와 SAD를 선형 결합

			EE1-5.1 Combination of the neural network-based intra prediction mode and ISP  JVET-AI0130 **Recommendations** • Good tradeoff, would be desirable to include. Training cross-check not finished, and likely expected not to finish before end of meeting. It was asked how the zero weights are distributed which are providing the complexity reduction. It is however pointed out that SADL takes care of this in a way commonly used for sparse networks. The cross check JVET-AI0273 was not finished before the closing of the meeting. If it is successful, JVET-AI0130 will be adopted in NNVC SW 10, and description v8 (to be confirmed in AHG telco).

❍ **Next EE [JVET-AI2023] [E. Alshina, R. Chang, F. Galpin, Y. Li, D. Rusanovskyy, M. Santamaria, J. Ström, and Z. Xie (EE coordinators)]**

- 금번 회의에서 진행한 EE activity review 및 non-EE 기고 review를 통해 차기 회의 (JVET-AJ)에서 진행할 EE를 결정함
- BoG (JVET-AI0339) 설정을 통해 실험을진행
- LOP 5건, HOP 3건, NN-inter 1건, NNSR 1건의 신규 EE 및 Next EE가 할당됨
- 차기 회의 EE를 요약한 테이블은 아래와 같음

EE1	Tests	Tester
1.1	Partial Convolution and Over-Parameterization (JVET-AI0068)	UESTC, Transsion
1.2	Simplified residual groups with attention (JVET-AI0221)	Qualcomm
1.3	NN in-loop filters using early cropping (JVET-AI0095)	Ericsson
1.4	Reduced complexity input feature extraction for LOP & VLOP (JVET-AI0204)	Qualcomm
1.5	LOP filter with multiscale blocks (JVET-AI0134)	N/A
2.1	Attention block with transformers (JVET-AI0176)	Qualcomm
2.2	Block-size invariant implementation of HOP (JVET-AI0175)	Qualcomm
2.3	Block based QP information in NN loop filters (JVET-AI0188)	InterDigital
3.1	RA/LDB Unified Reference Frame Synthesis for VVC Inter Coding (JVET-AI0089)	Xidian Univ.
4.1	Wavelet transform for super-resolution loss function (JVET-AI0105)	HUST

2.2. AHG12 Enhanced Compression Beyond VVC Capability

2.2.1. 회의 내용 및 결과

❍ **AHG report 검토 (M. Karczewicz, Y. Ye, L. Zhang (co-chairs), B. Bross, R. Chernyak, X. Li, K. Naser, H. Yang (vice-chairs))**

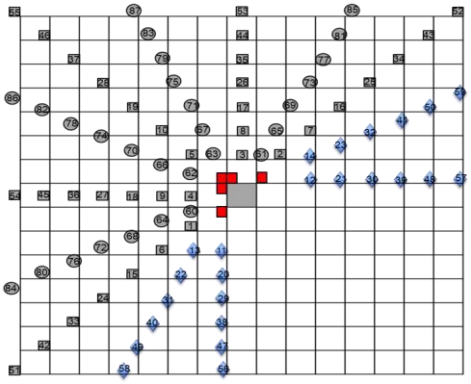
- Mandates
  - Solicit and study non-neural-network video coding tools with enhanced compression capabilities beyond VVC.
  - Discuss and propose refinements to the ECM13 algorithm description JVET-AH2025.
  - Coordinate with AHG7 to study the performance and complexity tradeoff of these video coding tools.
  - Coordinate with AHG6 on ECM software development.
  - Analyse the results of exploration experiments described in JVET-AH2024 in coordination with the EE coordinators.
  - Coordinate with AHG11 to study the interaction with neural network-based coding tools.
  - Define testing conditions and prepare bitstreams for subjective evaluation of ECM by expert viewing at the next meeting in coordination with AG 5, AHG4, and AHG6.

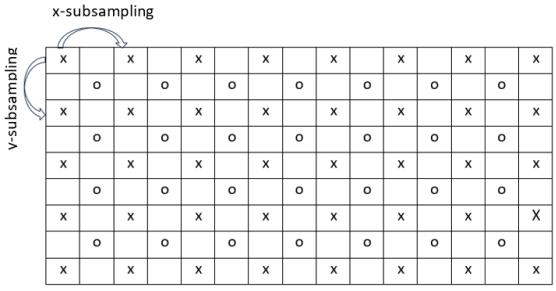
❍ **EE (Exploration experiment) Activity 검토**

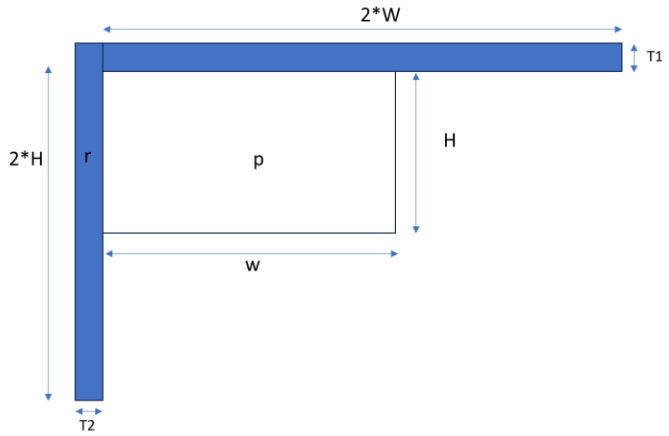
- 지난 34차 JVET 회의에서 제안된 부호화 툴에 대한 검토 후 6개의 EE를 구성하였으며, 이번 35차 JVET 회의에 제출된 EE 실험결과 (압축성능, 부/복호화기 복잡도, 기존 Tools간 interaction)을 검토하여 ECM-14.0 SW 채택 여부를 결정하였음.

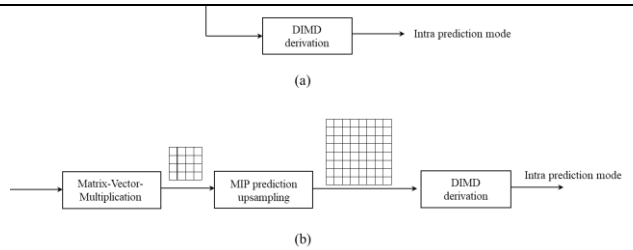
- **Intra Prediction (9건)**

EE2	Title	Coordinators	Reviews
2.1	TIMD merge	M. Abdoli (Xiaomi)	**Summary** • TIMD를 수행하는 과정에서 merge mode를 적용한 방법이 기고됨 • (Prediction modes, Fusion flag, Fusion weights, Wide-angle, MTS transform types)으로 구성된 리스트 생성 • TIMD merge list의 후보는 5개이며, 인접 및 비인접 블록에서 후보를 찾음

			 • Experimental results {Y, U, V, EncT, DecT} ➢ AI: -0.07%, 0.05%, -0.01%, 100.9%, 100.3% ➢ RA: -0.03%, 0.05%, -0.04% **Decision** **• Investigate in next EE**
2.2	TIMD with enhanced block vectors deployment	K. Naser (InterDigital) J. Fu (Peking Univ.)	**Summary** • TIMD의 fusion 과정 중 BV를 사용하는 방법이 기고됨 • Fusion 시 사용되는 non-angular mode를 BV merge list내의 후보를 사용함 • 이때 AR-BVP를 적용하여 추가 후보를 탐색함 • Experimental results {Y, U, V, EncT, DecT} ➢ AI: -0.04%, 0.00%, -0.06%, 100.4%, 101.1% ➢ RA: -0.02%, 0.00%, -0.13%, 99.6%, 98.6% **Decision** **• Investigate in next EE**
2.3	Combination of TIMD related tests	M. Abdoli (Xiaomi) K. Naser (InterDigital) J. Fu (Peking Univ.)	**Summary** • Test 2.1 / Test 2.5 / Test 2.3의 조합 실험 • Experimental results {Y, U, V, EncT, DecT} ➢ AI: -0.12%, 0.02%, 0.00%, 101.0%, 101.1% ➢ RA: 0.06%, -0.03%, 0.02%, 10x.x%, 10x.x% **Decision** **• No action**
2.4	Additional TIMD mode with different cost metric	D. Bugdayci Sansli (Nokia)	**Summary** • TIMD의 모드를 도출하는 cost function을 SATD 에서 MR-SAD로 변경한 방법이 기고됨 • 또한 Template size를 4에서 8로 변경함 • Test 2.4a: Additional TIMD mode with different cost metric • Test 2.4b: Test 2.4a without template size change • Experimental results {Y, U, V, EncT, DecT} ➢ 2.4a AI: -0.06, % 0.03, % 0.01, 100.6%, 100.9%,100.1%,100.0% RA: -0.03%, -0.05 %, 0.05 %, 100.7%, 99.1%, 100.2%, 100.1%

			➢ 2.4b AI: -0.03 %, 0.02 %, 0.03 % **Decision** **• A Investigate in next EE test 2.4a**
2.5	Intra-merge mode	Y.-J. Chang (Qualcomm)	**Summary** • 현재 블록을 부호화하기 위해 인접블록의 Intra mode를 사용하는 방법이 기고됨 • 인접 블록의 DIMD, TIMD 모드를 사용함 • Merge candidate list는 최대 4개의 후보로 구성되며, template cost를 비교하여 reordering을 수행함 • Intra merge mode 사용 여부를 위한 flag가 전송됨 • Test 2.5a: Intra merge mode • Test 2.5b: Test 2.5a without TIMD candidate • Test 2.5c: Test 2.5a with neighbour blocks pattern as in ECM intra tools • Test 2.5d: Test 2.5a with larger candidate list (3 candidates). • Experimental results {Y, U, V, EncT, DecT} ➢Test EE2-2.5a: Intra-merge mode (1 candidate): AI { -0.07%, 0.02%, 0.01%, 103.7%, 101.5%} RA { -0.04%, 0.00%, 0.00%, 100.0%, 100.0%} ➢Test EE2-2.5b: Intra-merge mode without TIMD candidates: AI { -0.06%, 0.00%, 0.00%, 103.5%, 101.6%} RA { 0.00%, -0.00%, 0.00%, 100.0%, 100.0%} ➢Test EE2-2.5c: Intra merge mode with adjacent and non-adjacent neighbors used in current ECM intra tools: AI { -0.06%, 0.02%, 0.02%, 103.6%, 101.2%} RA { 0.00%, -0.00%, 0.00%, 100.0%, 100.0%} ➢Test EE2-2.5d: Intra merge mode with larger candidate list (3 candidates): AI { -0.14%, 0.02%, 0.02%, 110.3%, 100.5%} RA { 0.00%, -0.00%, 0.00%, 100.0%, 100.0%} **Decision** **• No action**
2.6	IntraTMP candidates with overlapping refinement window enhanced	D. Ruiz Coll (Ofinno) K. Naser (InterDigital)	**Summary** • IntraTMP의 refinement search를 수행할 때 Merge candidate의 window 사이즈 (11x11)와 AR-BVP 및 Sparse search의 window 사이즈 (3x3)이 overlap 되는 경우를 해결하기 위한 방법이 기고됨 • Overlap 되는 경우를 Template cost가 낮은 BV후보를 사용함 • LIC의 탐색영역을 Half pel 만큼 shift함  • Experimental results {Y, U, V, EncT, DecT}

			➢ AI: -0.10%, -0.08%, -0.05%, 100.7%, 101.6% ➢ RA: -0.03%, 0.04%, -0.07%, 100.8%, 99.1% **Decision** **• Adopt, Test 2.6**
2.7	Regularized EIP/CCCM	H. Qin (TCL)	**Summary** • EIP mode의 성능을 높이기 위해 Regularization term을 수정한 방법이 기고됨 • Experimental results {Y, U, V, EncT, DecT} ➢ AI: -0.05%, -0.01%, 0.00%, 99.5%, 99.4% ➢ RA: -0.03%, -0.01%, -0.01%, 99.9%, 100.0% **Decision** **• Adopt, Test 2.7**
2.8	Modifications to matrix-based intra prediction	H. Wang (Qualcomm)	**Summary** • MIP의 성능을 높이기 위해 기존 MIP 방법이 아닌 template 영역을 이용하여 predictor를 생성하는 방법이 기고됨 • Last meeting (JVET-34th)에서 adoption된 intra mode 대체 방법을 modification한 방법임 • Original MIP의 memory usage는 본 기고보다 약 2배 낮음  • Experimental results {Y, U, V, EncT, DecT} ➢ AI: -0.08%, -0.03%, -0.06%, 100.5%, 100.5% ➢ RA: -0.04%, 0.01%, -0.01%, 100.0%, 99.8% **Decision** **• Adopt, Test 2.8**
2.9	Derived MIP modes with fusion	Z. Xie (OPPO)	**Summary** • MIP 정보를 template cost를 이용해 derivation하는 방법이 기고됨 • Template 영역을 MIP로 복원한 뒤 SATD를 비교하여 최대 2개의 후보를 선택하며, 첫번째 후보의 SATD 가 두번째 SATD보다 2배 이상 작지 않을 경우 fusion을 진행함

			 • Test 2.9a: MIP modes with fusion • Test 2.9b: MIP-LFNST transform set derivation by upsampled prediction • Test 2.9c: Test 2.9a + Test 2.9b • 비슷한 방법인 EE2-2.8과 비교하여 좋지 않은 trade-off를 가짐 • Experimental results of Test 2.9a{Y, U, V, EncT, DecT} ➢ AI: -0.04%, -0.03%, -0.04%, 100.5%, 99.8% ➢ RA: -0.04%, 0.05%, -0.08%, 100.3%, 100.0% • Experimental results of Test 2.9b{Y, U, V, EncT, DecT} ➢ AI: -0.01%, 0.01%, -0.05%, 99.8%, 100.0% ➢ RA: 0.00%, -0.04%, 0.02%, 100.1%, 100.1% • Experimental results of Test 2.9c{Y, U, V, EncT, DecT} ➢ AI: -0.05%, 0.03%, -0.03%, 100.5%, 100.0% ➢ RA: -0.04%, 0.01%, 0.05%, 100.4%, 99.6% **Decision** **• No action**

- **Inter Prediction (10건)**

EE2	Title	Contributors	Reviews
3.1	EE2-3.1: High Precision Sample Based Bi-Directional Optical Flow (JVET-AI0046)	Mehdi Salehifar, Yuwen He, Kai Zhang, Li Zhang (Bytedance)	**Summary** • 각 픽셀 단위로 MV를 추정해 refinement를 수행하는 Sample-based BDOF의 수식 변경 • 변경된 수식은 아래와 같음 (∑Gx.Gx+R1) * vx + ∑Gx.Gy * vy = ∑dI . Gx - dM . ∑Gx ∑Gx.Gy * vx + (∑Gy.Gy+R1) * vy = ∑dI . Gy - dM . ∑Gy 여기서 Gx 및 Gy는 각각 2개의 참조 블록에서 수평, 수직 방향의 gradient 합을 나타냄 • Test 3.1a: High accuracy sample based BDOF ✓RA: -0.10%, -0.09%, -0.08%, EncT: 101.1%, DecT: 100.5% • Test 3.2b: High accuracy sample based BDOF with no weight ✓RA: -0.03%, -0.03%, -0.00%, EncT: 101.0%, DecT: 100.6% **Decision**

<table>
<tr><td></td><td></td><td></td><td>• Adopt JVET-AI0046 test 3.1a</td></tr>
<tr><td>3.2</td><td>EE2-3.2: Temporal BV for IBC merge/AMVP list construction
(JVET-AI0161)</td><td>Luhang Xu, Yue Yu, Haoping Yu, Dong Wang (OPPO)
Na Zhang, Kai Zhang, Li Zhang (Bytedance Inc.)</td><td>Summary
• IBC merge 및 IBC AMVP에서 BV list 구성 과정에 temporal BVP를 추가하는 방법을 제안함
• 제안 방법은 아래 2가지 방법을 통해 temporal BVP를 추가함
1) Collocated positions: Col-block에 존재하는 5개 포인트를 지정하고 해당하는 위치에 BV가 존재하는 경우 BV list에 추가함

2) Shifted collocated positions: 현재 블록의 주변 또는 col-block에 존재하는 MV 위치만큼 shift한 영역에서 BV가 존재하는 경우 BV list에 추가함
- Col-block의 MV로 shift를 수행하는 경우 time scaling 적용
- 현재 블록의 주변 MV는 5개 중 1개를 선택해 사용함 (Fig.2 예시)

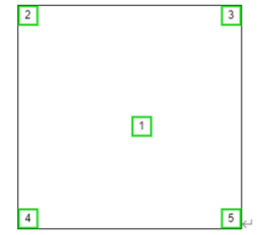
 

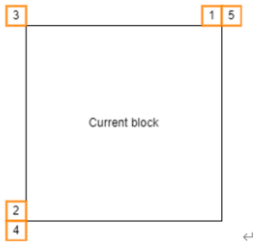
Figure-1. Five collocated positions.
 

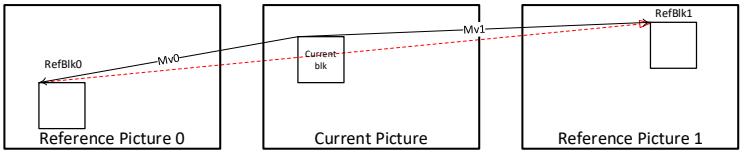
Figure-2. Five spatial neighboring positions.

• Random Access
✓ClassF: -0.03%, 0.02%, -0.07%, EncT 100.0%, DecT 100.0%
✓ClassTGM: -0.58%, -0.42%, -0.56%, EncT 99.5%, DecT 101.3%
• Low Delay B
✓ClassF: -0.24%, -0.32%, -0.69%, EncT 99.9%, DecT 99.5%
✓ClassTGM: -0.59%, -0.54%, -0.55%, EncT 100.2%, DecT 98.5%

Decision
• Test 3.3a와의 조합실험에서 adoption (Test 3.3b)</td></tr>
<tr><td>3.3</td><td>EE2-3.3: GPM with inter prediction and IBC
(JVET-AI0082)</td><td>Yang Wang, Na Zhang, Kai Zhang, Li Zhang (Bytedance)

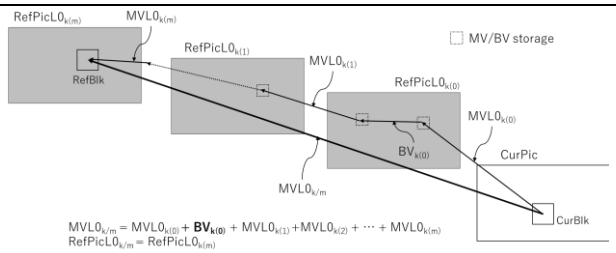
Luhang Xu, Yue Yu, Haoping Yu, Dong Wang (OPPO)</td><td>Summary
• GPM에서 inter (merge) predictor와 IBC predictor의 조합을 허용함
• IBC predictor는 BV list 중 하나를 선택하며, index를 명시적으로 복호화기에 전송함
• IBC predictor는 regular GPM, GPM-MMVD, GPM-affine에서 하나의 조합으로 사용할 수 있음
• Test 3.3b는 Test3.2와의 조합 실험 결과를 제시함

• Test results of Test 3.3a: 제안방법
<table><tr><td></td><td colspan="7">Random Access Main 10</td></tr><tr><td></td><td>Y</td><td>U</td><td>V</td><td>EncT</td><td>DecT</td><td>EncVmPeak</td><td>DecVmPeak</td></tr><tr><td>Class F</td><td>-0.02%</td><td>0.00%</td><td>-0.15%</td><td>100.40%</td><td>100.80%</td><td>100.20%</td><td>100.30%</td></tr><tr><td>Class TGM</td><td>-0.51%</td><td>-0.35%</td><td>-0.39%</td><td>99.80%</td><td>100.50%</td><td>100.60%</td><td>100.40%</td></tr></table>
<table><tr><td></td><td colspan="7">Low delay B Main 10</td></tr><tr><td></td><td>Y</td><td>U</td><td>V</td><td>EncT</td><td>DecT</td><td>EncVmPeak</td><td>DecVmPeak</td></tr><tr><td>Class F</td><td>-0.02%</td><td>0.29%</td><td>0.27%</td><td>100.10%</td><td>100.20%</td><td>100.70%</td><td>99.70%</td></tr><tr><td>Class TGM</td><td>-0.71%</td><td>-0.51%</td><td>-0.62%</td><td>100.00%</td><td>99.90%</td><td>100.50%</td><td>101.00%</td></tr></table></td></tr>
</table>

<table>
<tr>
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<td></td>
<td></td>
<td>
• Test results of Test 3.3b: 제안방법 + Test 3.2
<table>
<tr><td></td><td colspan="7">Random Access Main 10</td></tr>
<tr><td></td><td>Y</td><td>U</td><td>V</td><td>EncT</td><td>DecT</td><td>EncVmPeak</td><td>DecVmPeak</td></tr>
<tr><td>Class F</td><td>-0.03%</td><td>0.05%</td><td>-0.04%</td><td>99.60%</td><td>99.40%</td><td>100.20%</td><td>99.90%</td></tr>
<tr><td>Class TGM</td><td>-1.07%</td><td>-0.92%</td><td>-0.98%</td><td>100.50%</td><td>100.80%</td><td>100.30%</td><td>99.30%</td></tr>
</table>
<table>
<tr><td></td><td colspan="7">Low delay B Main 10</td></tr>
<tr><td></td><td>Y</td><td>U</td><td>V</td><td>EncT</td><td>DecT</td><td>EncVmPeak</td><td>DecVmPeak</td></tr>
<tr><td>Class F</td><td>-0.27%</td><td>-0.24%</td><td>-0.31%</td><td>99.90%</td><td>99.70%</td><td>100.20%</td><td>100.00%</td></tr>
<tr><td>Class TGM</td><td>-1.50%</td><td>-1.44%</td><td>-1.64%</td><td>99.40%</td><td>100.70%</td><td>100.70%</td><td>97.40%</td></tr>
</table>
Decision
• Adopt JVET-AI0082 test 3.3b (only for CTC in SCC)
</td>
</tr>
<tr>
<td>3.4</td>
<td>EE2-3.4: MVP extension
(JVET-AI0183)</td>
<td>Zhi Zhang, Jian-Liang Lin, Han Huang, Vadim Seregin, Marta Karczewicz (Qualcomm)</td>
<td>
Summary
• Merge list 구성 방법 중 TMVP에 대한 확장 방법 3가지를 제안함
• TMVP 확장방법 3가지는 각각 아래와 같음
1) Scaled merge candidates: Reference index 0으로 scaling 된 TMVP가 존재한다면, Reference index 1로 scaling 한 TMVP도 merge list에 추가함
2) Additional TMVP and SbTMVP: Col-block이 inter로 부/복호화가 된 경우, col-block이 가리키는 참조 프레임을 기준으로 time scaling을 수행하여 merge list에 추가함 (SbTMVP에도 적용됨)
3) Additional bi-TMVP candidates: Uni MV가 가리키는 참조 블록에서 사용한 참조 픽처가 현재 블록을 교차한다면 (아래 그림 중 빨간색 화살표 예시) 참조 블록이 가지는 MV를 이용해 Bi MV로 이용하여 merge list에 추가함


• 상기 3가지 방법과 조합한 방법에 대한 실험 결과를 제시함
• Test results of Test 3.4a: Scaled merge candidates
- RA: -0.01%, 0.01%, -0.07%, 99.5%, 99.3%
• Test results of Test 3.4b: Additional TMVP and SbTMVP
- RA: -0.06%, -0.10%, -0.07%, 100.4%, 100.1%
• Test results of Test 3.4c: Additional bi-TMVP candidates
- RA: -0.04%, -0.07%, -0.05%, 99.6%, 100.3%
• Test results of Test 3.4d: 3.4a + 3.4b + 3.4c
- RA: -0.11%, -0.08%, -0.08%, 100.5%, 100.6%
• Test results of Test 3.4e: 3.4b + 3.4c
- RA: -0.09%, -0.08%, -0.14%, 100.6%, 100.4%
Decision
• 3.4a에 대한 benefit이 거의 없다고 판단되며, 3.4d와 3.4e의 복잡도 차이는 신뢰할 수 없다고 판단함
• Adopt JVET-AI0183 test 3.4e (only for CTC in RA)
</td>
</tr>
<tr>
<td>3.5</td>
<td>EE2-3.5: Additional chained motion vector</td>
<td>Na Zhang, Kai Zhang, Li Zhang (Bytedance)</td>
<td>
Summary
• Test 3.5a: PAMVP 구성을 수행할 후보에 대해 CMVP 후보를 추가로 사용하도록 변경함
- Low delay picture에서는 PAMVP 개수를 6까지 사용가능
</td>
</tr>
</table>

	prediction candidates (JVET-AI0103)		- Non low delay picture에서는 PAMVP 개수를 2까지 사용 가능 • Test 3.5b: Zero vector로 채우기 이전에 CMVP 후보를 최대 4개 추가로 merge list에 삽입 - 원래 CMVP를 추가하는 부분은 변경하지 않고 Zero vector 이전에 추가만 수행함 • Test results of Test 3.5a: PAMVP 구성 중 CMVP 추가 - RA: -0.01%, -0.06%, -0.01%, 99.9%, 100.0% - LDB: -0.10%, 0.07%, -0.02%, 99.9%, 101.0% • Test results of Test 3.5b: Zero vector 이전 CMVP 추가 - RA: -0.04%, -0.05%, -0.08%, 100.2%, 100.9% - LDB: -0.08%, -0.03%, -0.42%, 100.3%, 101.9% **Decision** • Adopt JVET-AI0103 test 3.5b
3.6	EE2-3.6: Adaptive cost function selection in merge mode (JVET-AI0185)	Kai Cui, Zhi Zhang, Yan Zhang, Han Huang, Vadim Seregin, Marta Karczewicz (Qualcomm)	**Summary** • Merge list 구성 이후 각 index 별 cost function을 서로 다르게 사용하는 방법에 대해 제안함 - Merge index가 짝수인 경우 1번 cost function을 사용함 - Merge index가 홀수인 경우 1번 cost function을 사용함 • Bilateral matching merge, Affine merge에서는 아래 cost function을 사용함 - 1번 cost function: SATD - 2번 cost function: SAD • Template matching merge 아래 cost function을 사용함 - 1번 cost function: Weighted SAD - 2번 cost function: SAD • Test results of Test 3.6 - RA: -0.06%, -0.02%, -0.02%, 100.1%, 100.4% - LDB: -0.08%, -0.21%, 0.02%, 99.7%, 99.4% **Decision** • Adopt JVET-AI0185 test 3.6
3.7	EE2-3.7: TMVP for chained motion vector prediction (JVET-AI0187)	P.-H. Lin, Z. Zhang, J.-L. Lin, V. Seregin, M. Karczewicz (Qualcomm)	**Summary** • CMVP 구성 중 첫번째 순서로 사용되는 MVP를 TMVP로도 사용할 수 있도록 변경 • 추가적으로 Bilateral Matching (BM) merge mode에서도 CMVP를 사용할 수 있도록 확장

			 <CMVP 구성 방법 예시> • Test results of Test 3.7 - RA: -0.07%, -0.09%, -0.08%, 99.8%, 100.3% - LDB: -0.02%, 0.13%, 0.03%, 99.7%, 100.1% **Decision** • Adopt JVET-AI0187 test 3.7
3.8	EE2-3.8: GPM-affine with MMVD and TM (JVET-AI0083)	Y. Wang, K. Zhang, L. Zhang (Bytedance)	**Summary** • GPM-affine 예측에서 MVP 보정을 위해 MMVD 또는 TM을 사용할 수 있도록 변경 • GPM-affine에 대해 MMVD 적용 (GPM-affine-MMVD), TM 적용 (GPM-affine-TM)에 대한 각각의 성능과 조합실험 성능을 함께 제시함 • Test results of Test 3.8a: GPM-affine with GPM-MMVD - RA: -0.03%, -0.07%, -0.03%, 101.5%, 99.8% • Test results of Test 3.8b: GPM-affine with GPM-TM - RA: -0.05%, -0.04%, -0.04%, 100.7%, 99.6% - LDB: -0.11%, -0.12%, -0.08%, 101.4%, 100.0% • Test results of Test 3.8c: Test 3.8a + Test 3.8b - RA: -0.06%, -0.09%, -0.05%, 102.0%, 99.9% - LDB: -0.11%, -0.12%, -0.08%, 101.7%, 99.9% **Decision** • Investigate test 3.8b in EE that GPM related proposals from non-EE category would be tested
3.9	EE2-3.9: Affine candidates derived from temporal collocated pictures (JVET-AI0197)	C. Ma, X. Xiu, C.-W. Kuo, W. Chen, X. Wang (Kwai)	**Summary** • Temporal collocated picture에 존재하는 CPMVP들을 이용해 현재 블록의 Affine candidate를 추가할 수 있도록 확장 • 해당 추가 candidate는 현재 블록의 position과 size에 의해 scaling을 수행하며, affine merge candidate로 추가된 이후, ARMC를 수행함 • Test results of Test 3.9 - RA: -0.03%, -0.03%, -0.02%, 100.4%, 100.9% - LDB: -0.30%, -0.37%, -0.20%, 100.9%, 102.9% **Decision** • Adopt JVET-AI0197 test 3.9

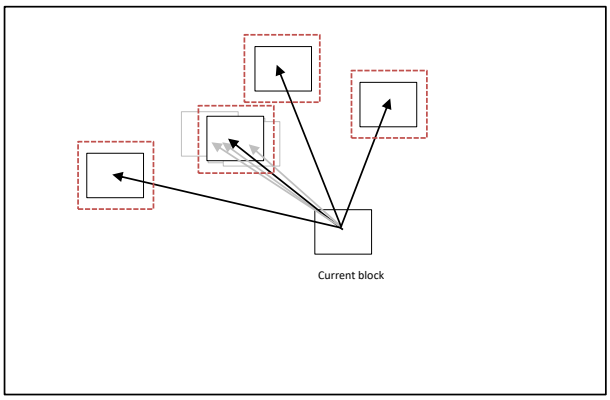
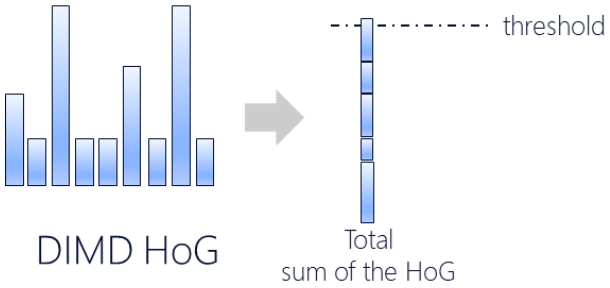
3.10	EE2-3.10 Sharp motion compensation filter for bi-prediction (JVET-AI0094)	J. Samuelsson-Allendes (Sharp)	**Summary** • Sub-pel interpolation을 위한 신규 interpolation filter 계수 제안 • 현재 ECM filter계수와 제안하는 filter 계수의 차이는 아래 그림과 같음 • Test 3.10b는 일부 bug-fix와 작은 block size (4x4)에 대해서도 제안하는 interpolation filter를 사용하도록 수정함 ecmFilter – proposedBiPredFilter = { { 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, { 0, -1, 2, -2, 2, -2, 1, -1, 1, -1, 1, 0}, { 1, -2, 2, -3, 4, -4, 3, -3, 3, -2, 1, 0}, { 1, -2, 3, -4, 4, -3, 4, -4, 3, -2, 1, -1}, { 1, -3, 4, -4, 5, -5, 4, -4, 3, -2, 2, -1}, { 1, -3, 3, -4, 5, -4, 4, -4, 3, -2, 1, 0}, { 1, -1, 2, -3, 3, -4, 3, -3, 3, -2, 1, 0}, { 0, -1, 1, -1, 2, -2, 1, -1, 2, -1, 1, -1}, { 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, { -1, 1, -1, 2, -1, 1, -2, 2, -1, 1, -1, 0}, { 0, 1, -2, 3, -3, 3, -4, 3, -3, 2, -1, 1}, { 0, 1, -2, 3, -4, 4, -4, 5, -4, 3, -3, 1}, { -1, 2, -2, 3, -4, 4, -5, 5, -4, 4, -3, 1}, { -1, 1, -2, 3, -4, 4, -3, 4, -4, 3, -2, 1}, { 0, 1, -2, 3, -3, 3, -4, 4, -3, 2, -2, 1}, { 0, 1, -1, 1, -1, 1, -2, 2, -2, 2, -1, 0}, • Test results of Test 3.10a - RA: -0.06%, -0.02%, -0.05%, 100.2%, 100.0% - LDB: -0.03%, 0.11%, -0.18%, 100.1%, 99.5% • Test results of Test 3.10b: Bug-fix 포함 - RA: -0.07%, 0.00%, -0.01%, 100.2%, 99.8% - LDB: -0.10%, -0.11%, -0.16%, 100.1%, 99.6% **Decision** • Adopt JVET-AI0094 test 3.10b

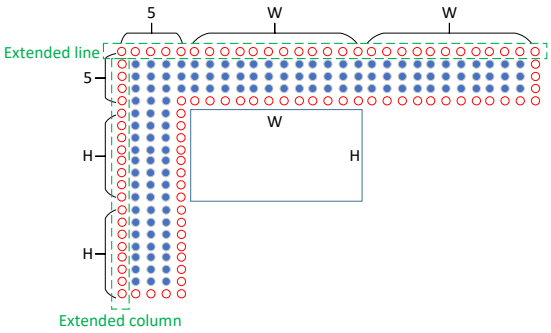
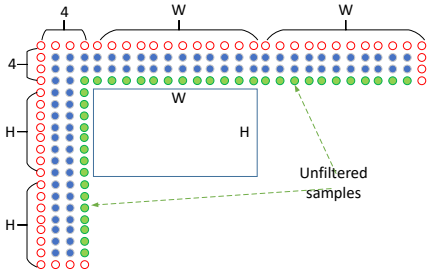
❍ 신규 기술 기고서 검토

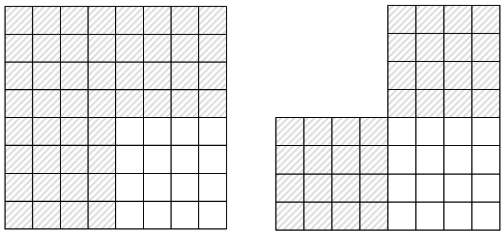
- 이번 35차 JVET 회의에서 새로운 부호화 향상 tool 제안에 대한 기술 기고서가 총 56건 (Intra 21건, Inter 12건, 기타 23건) 제출됨.
- 아래 표에 주요 기술 기고서에 관한 논의사항 및 결정사항을 요약 정리하였으며, 각 기고서의 상세한 기술적 방법은 추후 실 내 세미나를 통해서 다룰 예정임.

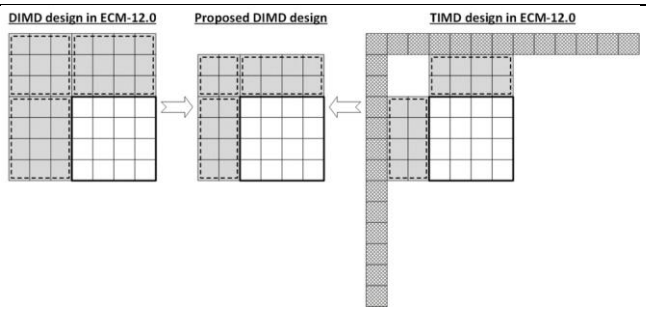
- **Intra Prediction (3 + 18건)**

Category	Title	Contributors	Reviews
Related	EE2 related: BV Merge List Improvement (JVET-AI0253)	K. Naser, Y. Chen, M. Radosavljević, F. Le Léannec (InterDigital)	**Summary** • Block vector merge list construction 과정에서 refinements window를 통해 block vector를 조정하는 방법이 기고됨 • 비슷한 위치를 가리키는 Block vector는 clustering을 통해 하나의 refinements window를 사용함

			 • Test result compares to ECM-13.0 anchor ✓AI: 0.09% -0.04% -0.07%, 101.5%, 102.5% **Decision** • Interesting gain – **study in EE**
Non-EE2	AHG12: Adaptive HoG for DIMD (JVET-AI0106)	S. Blasi, J. Lainema (Nokia)	**Summary** • DIMD의 HoG를 추출하는 과정에서 고정된 Template영역을 사용하는 것이 아닌 적응적으로 sample수를 결정하는 방법이 기고됨 • 총 누적 HoG값이 threshold 값을 넘을 때 까지 sobel filter를 적용함 • 멀리 있는 영역에 sobel filter를 적용하는 경우 scaling process를 적용함 (최대 32 line)  • Test result compares to ECM-13.0 anchor ✓AI: -0.04%, -0.03%, -0.02%, 100.1%, 102.8%, 100.5%, 100.2% **Decision** • **Study in EE** • 현재 결과보다 낮은 Decoding time을 요구함 (현재 102.8%)
Non-EE2	AHG12: Adaptive (JVET-AI0108)	N. Zhang, K. Zhang, Y. Wang, Z. Deng, L. Zhao, W. Yin,	**Summary** • IntraTMP와 IBC의 search range를 통합하기 위해 IBC의 search range를 확장하는 방법이 기고됨 • Search range 통합을 위해 IBC의 탐색 가능 영역을 현재 frame 내의 모든 영역으로 확장함

		L. Zhang (Bytedance)	• Test result compares to ECM-13.0 anchor ✓ AI: Overall: {0.00%, -0.02%, 0.00%, 99.2%, 99.5%}; Class F: {-0.29%, -0.22%, -0.17%, 98.7%, 98.1%}; Class TGM: {-1.22%, -1.28%, -1.27%, 99.4%, 96.0%}; ✓ RA: Overall: {-0.01%, -0.04%, 0.00%, 99.2%, 99.5%}; Class F: {-0.18%, -0.04%, -0.28%, 99.4%, 99.4%}; Class TGM: {-0.76%, -0.80%, -0.71%, 99.6%, 98.9%}. **Decision** **• Study in EE**
Non-EE2	Non-EE2: DIMD with filtered-template (JVET-AI0125)	Z. Lv, C. Zhou (vivo)	**Summary** • DIMD의 HoG추출 과정 중 filtered template를 사용하는 방법이 기고됨 • Filtering 수행 시 좌측 및 상단에 2개의 line을 추가하여 3x3 filtering을 수행함 $$F = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$ • 추가 2개의 line이 모두 가용한 경우 template 영역에 filtering을 수행하여 사용함 (Figure 1) • 추가 1개의 line만 사용 가능한 경우 0번 line은 filtering되지 않은 sample을 사용함 (Figure 2)  Figure 1. Template-Ⅰ  Figure 2. Template-Ⅱ

			• Test result compares to ECM-13.0 anchor ✓AI: -0.05%, 0.00%, -0.03% **Decision** **• Investigate in EE** • 좋은 trade-off를 위해서 1% 이하의 encoding runtime을 요구함
Non-EE2	Non-EE2: Modified L-shape template in IntraTMP (JVET-AI0139)	J. Lee, J. Kim, W. Lim, D. Kim, S.-C. Lim (ETRI)	**Summary** • IntraTMP의 Template 영역을 수정하는 방법이 기고됨 • 현재 block을 기준으로 좌측 상단 영역의 template은 현재 block과 연관성이 크지 않다고 판단하여 제거함  (a) ECM-13.0 (b) Proposal Figure 1. An example of L-shape template for 4x4 block • Test result compares to ECM-13.0 anchor ✓AI: -0.03%, 0.03%, -0.04%, 99.6%, 100.1% ✓RA: -0.03%, -0.01%, -0.01%, 100.1%, 99.7% **Decision** **• Investigate in EE**
Non-EE2	Non-EE2: On DIMD edge operators (JVET-AI0140)	A. Filippov, V. Rufitskiy, J. Konieczny, H. Qin, T. Dong, Z. Xu (TCL)	**Summary** • DIMD의 HoG추출 과정 중 sobel filter의 크기를 변경하는 방법이 기고됨 • 크기가 작은 block (4x4, 4x8, 8x4) 에 대해서 기존 3x3 filter 에서 2x2 로 변경함 $$F_x = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$$ $$F_y = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

<table>
<tr>
<td></td>
<td></td>
<td></td>
<td> 

• Test result compares to ECM-13.0 anchor
✓AI: -0.02%, -0.01%, 0.00%, 99.2%, 100.1%

Decision
• Investigate in EE</td>
</tr>
<tr>
<td>Non-EE2</td>
<td>AHG12: neural network-based intra prediction (JVET-AI0201</td>
<td>F. Urban, T. Dumas, F. Galpin, E. François (InterDigital)</td>
<td>Summary
• NNVC에서 사용하는 NN-intra를 ECM에 적용한 방법을 기고함
• Table 1은 각 모델에 필요한 parameter수와 저장 메모리를 나타냄

Table 1: parameters count and storage memory needed for each model.

<table>
<tr><th>model</th><th>parameters count</th><th>non-zero parameters count</th><th>memory footprint</th></tr>
<tr><td>4x4</td><td>968k</td><td>77k</td><td>155 kB</td></tr>
<tr><td>8x4</td><td>1008k</td><td>121k</td><td>242 kB</td></tr>
<tr><td>16x4</td><td>1888k</td><td>182k</td><td>365 kB</td></tr>
<tr><td>32x4</td><td>2122k</td><td>229k</td><td>458 kB</td></tr>
<tr><td>8x8</td><td>2122k</td><td>194k</td><td>388 kB</td></tr>
<tr><td>16x8</td><td>2317k</td><td>239k</td><td>477 kB</td></tr>
<tr><td>16x16</td><td>1699k</td><td>291k</td><td>583 kB</td></tr>
</table>

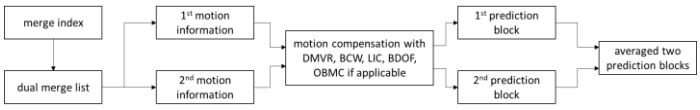
• Test result compares to ECM-13.0 anchor
✓AI: -0.60%, -0.43%, -0.46%, 107.0%, 119.5%

• 직전회의 (JVET 34th) 의 결과와 다르게 gain-runtime trade-off가 허용가능한 수준이라고 판단됨
• 하지만 Decoder runtime이 다른 기고서와 비교하여 높음

Decision
• Investigate in EE
• Training script 또한 같이 요구함</td>
</tr>
</table>

- **Inter Prediction (3 + 9건)**

Category	Title	Contributors	Reviews

Related	EE2-related: on the sharp motion compensation filter for bi-prediction (JVET-AI0217)	Han Huang, Ruoyang Yu, Vadim Seregin, Marta Karczewicz (Qualcomm)	**Summary** • EE2 Test 3.10에서 제안된 sharp motion compensation을 이용한 bi-prediction 과정 중 bug-fix 수행 • 현재 EE 소프트웨어 상 제안된 신규 interpolation filter 계수는 affine TM, affine merge에서는 사용되고 있지 않으나, 본 기고에서 신규 interpolation filter 계수를 모두 사용하도록 변경함 • Test result compares to ECM-13.0 anchor ✓RA: -0.08%, -0.01%, 0.04%, 98.9%, 98.7% ✓LDB: -0.06%, 0.12%, -0.27%, 99.3%, 98.4% **Decision** • Interesting gain – **study in EE**
Non-EE2	Non-EE2: A dual merge mode (JVET-AI0109)	Ru-Ling Liao, Jie Chen, Yan Ye, Xinwei Li (Alibaba)	**Summary** • Dual merge list를 구성해 최대 4개의 Motion information을 이용한 predictor 생성 방법 제안 • Dual merge list 구성 방법은 아래와 같음 1) Spatial, temporal, NA spatial block 중 dual merge로 부/복호화를 수행한 경우 dual merge list에 추가 2) 상기 방법으로 dual merge list를 모두 채우지 못한 경우, regular merge list에 존재하는 2개의 motion information 조합을 통해 dual merge list에 추가 3) ARMC 수행 • Dual merge 사용을 위해 기존 merge index와 dual merge flag 전송 • Dual merge를 수행한 블록은 2개의 MV를 저장해야 하므로, 기존 merge에 비해 MV를 저장하기 위한 메모리 대비 2배를 요구함  <Dual merge 예측 블록 생성 방법> • Test result compares to ECM-13.0 anchor ✓RA: -0.04%, -0.01%, -0.06%, 102.8%, 105.0% ✓LDB: -0.07%, 0.01%, -0.31%, 101.8%, 101.2% **Decision** • **Study in EE**, significant reduction in run time should be achieved
Non-EE2	Non-EE2: SATD-based reordering (JVET-AI0114)	Yang Wang, Kai Zhang, Wenbin Yin, Zhipin Deng, Na Zhang, Lei Zhao, Li Zhang (Bytedance)	**Summary** • Template matching 기반 reordering에 사용되는 template cost를 기존 SAD에서 SATD로 변경 • SATD 계산을 위한 template은 1개 line만 사용함 • Aspect #1: Intra tool에 대한 결과 제시 (TMRL, EIP, IBC 등) • Aspect #2: Inter tool에 대한 결과 제시 (ARMC)

			• Test result compares to ECM-13.0 anchor: Aspect #1 ✓AI: -0.02%, 0.00%, -0.03%, 100.2%, 99.8% ✓RA: -0.01%, 0.00%, -0.05%, 98.0%, 99.8% • Test result compares to ECM-13.0 anchor: Aspect #2 ✓RA: -0.03%, 0.03%, -0.08%, 100.7%, 100.4% **Decision** • SAD → SATD 변경에 따른 디코더 복잡도 증가가 없는 이유에 대해 문의했으나, 정확한 답변을 받지 못함 • **Investigate in EE**, the two aspects should be studied separately and also in combination with each other
Non-EE2	Non-EE2: Enhanced inter AMVP (JVET-AI0126)	Zhipin Deng, Kai Zhang, Na Zhang, Yang Wang, Wenbin Yin, Lei Zhao, Li Zhang (Bytedance)	**Summary** • 현재 ECM merge list 구성 방법에 사용하는 NA-SMVP를 AMVP list 구성에서도 사용할 수 있도록 수정 • Affine AMVP list에 대한 template 기반 정렬 추가 (최대 10개 후보를 가질 수 있음) • Test result compares to ECM-13.0 anchor ✓RA: -0.03%, -0.03%, -0.04%, 100.7%, 100.4% ✓LDB: -0.12%, -0.07%, -0.14%, 101.2%, 100.8% **Decision** • **Investigate in EE**, The two elements should be separately and the combination
Non-EE2	Non-EE2: On sub-block merge mode (JVET-AI0169)	Fan Wang, Yue Yu, Haoping Yu, Dong Wang, (OPPO)	**Summary** • Aspect #1: 기존 ECM에 적용된 sub-block merge mode (affine merge 포함)의 block size 제약 조건을 변경함 - 기존 제약 조건: width >= 8 && height >= 8 - 신규 제약 조건: width * height >= 32 • 따라서, 추가로 sub-block merge를 수행할 수 있는 블록 크기는 아래와 같음 - 4x8, 4x16, 4x32 … and 8x4, 16x4, 32x4 • Aspect #2 (Aspect #1 포함): 주변 블록의 정보를 이용해 현재 블록에서 sub-block merge를 수행할 것인지 결정하는 방법을 제안함 • 인접한 블록과 비인접한 블록 중 sub-block merge 혹은 affine AMVP가 1개라도 사용된 경우에만 현재 블록에 대한 sub-block merge를 수행함 • 만약 현재 블록에 대한 sub-block merge mode가 제한된 경우, sub-block merge flag에 대한 전송을 생략함 • Test result compares to ECM-13.0 anchor: Aspect #1 ✓RA: -0.08%, -0.07%, -0.05% • Test result compares to ECM-13.0 anchor: Aspect #2 ✓RA: -0.09%, -0.09%, -0.07%, 101.2%, 100.3%

			Decision • **Investigate in EE** • Aspect #2는 encoder에서만 사용도 가능한 방법으로 판단되어, encoder-only modification에 대해서도 실험을 수행할 것을 권고함
Non-EE2	Non-EE2: On POC conditions of BDOF (JVET-AI0198)	Ning Yan, Xiaoyu Xiu, Wei Chen, Xianglin Wang (Kwai)	**Summary** • BDOF 적용 조건 중 하나인 양방향 참조 프레임간 거리 (POC difference)가 동일한 경우 뿐만 아니라, 각 참조 프레임이 한 방향에만 존재하더라도 BDOF를 적용할 수 있도록 변경 • 양방향 참조 프레임에서 POC difference가 동일해야한다는 전제는 유지 • 한 방향에서 2개의 참조 프레임을 사용하는 경우, POC difference가 서로 다르더라도 BDOF 적용 가능 • Test result compares to ECM-13.0 anchor ✓RA: 0.00%, -0.01%, 0.02%, 100.0%, 99.9% ✓LDB: -0.12%, -0.02%, -0.15%, 101.1%, 101.6% **Decision** • **Investigate in EE** • 금번 회의에서 기고된 high precision BDOF와도 함께 조합 실험을 수행할 것을 권고함
Non-EE2	Non-EE2: On subblock inter mode improvement (JVET-AI0218)	Han Huang, Yan Zhang, Jian-Liang Lin, Zhi Zhang, Po-Han Lin, Chun-Chi Chen, Vadim Seregin, Marta Karczewicz (Qualcomm)	**Summary** • Subblock inter mode (Affine AMVP 포함)에 대한 3가지 수정을 제안함 1) Affine DMVR에서 CPMV refinement 과정 추가 2) CPMV를 기반으로 한 sub-block motion 추정 과정에서 sub-pel 정밀도를 기존 1/16-pel에서 1/64-pel 정밀도로 변경 (sub-block motion 저장은 1/16-pel로 다시 scaling 하여 저장, 해당 변경은 Luma에 대해서만 적용) 3) CPMV candidate list 구성 과정에서 TMVP 또는 CMVP를 결합한 MVP를 candidate에 추가하도록 확장 • Test result compares to ECM-13.0 anchor ✓RA: -0.12%, -0.11%, -0.16%, 101.5%, 102.9% ✓LDB: -0.21%, -0.22%, -0.21%, 102.0%, 101.8% **Decision** • **Investigate in EE** • 3가지 방법에 대한 독립 성능을 함께 확인을 권고함

❍ 주요 결정사항

- **[1] ECM 14.0 채택 기술**

Partioning
① [JVET-AI0136] Test 1.1c: EE2-1.1: adaptive dual tree in inter Slices
② [JVET-AI0087] Test 1.2b: EE2-1.2: Restricting BT CUs to apply QT-like partitioning structure

Intra prediction
① [JVET-AI0129] Test 2.6c: EE2-2.6: IntraTMP candidates with overlapping refinement window enhanced
② [JVET-AI0066] Test 2.7a: EE2-2.7: Regularized EIP/CCCM
③ [JVET-AI0208] Test 2.8a: EE2-2.8: Modifications to Matrix-based intra prediction
Inter prediction
① [JVET-AI0046] Test 3.1a: EE2-3.1: High Precision Sample Based Bi-Directional Optical Flow
② [JVET-AI0082] Test 3.3b (Test 3.2포함): EE2-3.3: GPM with inter prediction and IBC
③ [JVET-AI0183] Test 3.4e: EE2-3.4: MVP extension
④ [JVET-AI0103] Test 3.5b: EE2-3.5: Additional chained motion vector prediction candidates
⑤ [JVET-AI0185] Test 3.6: EE2-3.6: Adaptive cost function selection in merge mode
⑥ [JVET-AI0187] Test 3.7: EE2-3.7: TMVP for chained motion vector prediction
⑦ [JVET-AI0197] Test 3.9: EE2-3.9: Affine candidates derived from temporal collocated pictures
⑧ [JVET-AI0094] Test 3.10b: EE2-3.10 Sharp motion compensation filter for bi-prediction
Transform and coefficient coding
① [JVET-AI0050] Test 4.1c: EE2-4.1: Improvements on Inter LFNST/NSPT
Loop filtering
① [JVET-AI0058] Test 5.1a: EE2-5.1: Modifications in ALF syntax
② [JVET-AI0084] Test 5.2: EE2-5.2: ALF residuals scaling
③ [JVET-AI0166] Test 5.3a: EE2-5.3: CCALF with Chroma inputs
Entropy coding
① N/A
CTC for EE2/ECM and ECM improvements
① [JVET-AI0096] AHG12: Fixes for Adaptive Clipping and Sign Prediction
② [JVET-AI0097] AHG12: A follow-up CTC modification for encoding runtime reduction under LD config
③ [JVET-AI0281] AHG12: Context retraining for ECM-13.0

- **[2] 신규 EEs 구성 (JVET-AI2024)**
  - 이번 35차 JVET 회의에서 ECM tool 성능 분석을 위한 6개의 EEs가 신규 생성되었으며, 이에 대한 성능개선 결과가 차기 회의에서 논의될 예정임.
  - 총 43건의 EEs 중 Intra prediction 및 Inter prediction은 각각 21건 및 11건이 생성됨.
  - Next EE List (Intra Prediction)

EE2	Tests	Tester
2.1	Matrix-based position dependent intra prediction for TIMD	Y. Kidani(KDDI)
2.2	Intra merge mode extension with BV improvement	J. Fu (Peking Univ.) Y.-J. Chang

		(Qualcomm)
2.3	TIMD with enhanced block vectors deployment	K. Naser (InterDigital) J. Fu (Peking Univ.)
2.4	Test 2.3 + BV merge list improvement	K. Naser (InterDigital)
2.5	TIMD merge mode	M. Abdoli (Xiaomi)
2.6	Additional TIMD mode with different cost metric	D. Bugdayci Sansli (Nokia)
2.7a 2.7b 2.7c 2.7d	Disabling TIMD related combinations in normative way Disabling TIMD related combinations in non-normative way Test 2.7a + additional RDOs Test 2.7b + additional RDOs	Z. Fan (Sharp)
2.9	DIMD with filtered-template	Z. Lv (vivo)
2.10a 2.10b 2.10c	Adaptive HoG for DIMD Adaptive HoG without increased predictors Adaptive HoG without scaling and increased predictors	S. Blasi (Nokia)
2.11	DIMD with 2x2 edge operator applied to small blocks	A. Filippov (TCL)
2.12	Non-adjacent DIMD mode derivation for TMRL intra mode candidate list construction	V. Rufitskiy (TCL)
2.13a 2.13b 2.13c	Test 2.11 + Test 2.10 Test 2.11 + Test 2.9 Test 2.11 + Test 2.12	A. Filippov (TCL) S. Blasi (Nokia) V. Rufitskiy (TCL) Z. Lv (vivo)
2.14	High-level control of intra prediction methods	K.-Y. Kim (WILUS)
2.15a 2.15b 2.15c	Regression-based SGPM blending SGPM with PDP replacement Test 2.15a + Test 2.15b	X. Li (Alibaba)
2.16	Unified reference area of IBC and IntraTMP	N. Zhang (Bytedance)
2.17	Modified L-shape template in IntraTMP	J. Lee (ETRI)
2.18	Chroma TMRL	H. Huang (OPPO)
2.19a 2.19b 2.19c	Multi-model EIP Block vector guided EIP Test 2.19a + Test 2.19b	L. Zhang (OPPO) Z. Xie (OPPO)
2.20	Neural network-based intra prediction with DIMD mode derivation	S. Eadie (Qualcomm)
2.21	Neural network-based intra prediction	T. Dumas (InterDigital)

- Next EE List (Inter Prediction)

EE2	Tests	Tester

3.1	Dual merge mode	R.-L. Liao (Alibaba)
3.2a 3.2b 3.2c	Regular inter AMVP list with more candidates Affine inter AMVP list with template matching Test 3.2a + Test 3.2b	Z. Deng (Bytedance)
3.3a 3.3b 3.3c 3.3d	OBMC extension with intra prediction Test 3.3a + DIMD with 2x2 edge operator Test 3.3a + using intra prediction mode of neighbouring block Test 3.3a + more neighbouring blocks are checked for available motion information to generate inter subblock	D. Kim (ETRI)
3.4a 3.4b 3.4c 3.4d 3.4e	GPM mode extension (64 to 112 modes) GPM mode extension with block shape dependency (64 to 80 modes) GPM mode extension with additional distances for hor/ver angles only (64 to 70 modes) Test 3.4b not applied on IBC-GPM nor SGPM GPM mode extension with block size adaptation	Y. Ahn (LGE) K. Reuzé (InterDigital)
3.5	On POC conditions of BDOF	N. Yan (Kwai)
3.6	GPM-affine with TM	Y. Wang (Bytedance)
3.7a 3.7b	Sub-block merge mode extension with normative conditioning Sub-block merge mode extension with non-normative conditioning	F. Wang (OPPO)
3.8	Sharp motion compensation filter extension	H. Huang (Qualcomm)
3.9a 3.9b 3.9c 3.9d	Affine CPMV search for affine DMVR High precision MV for affine motion compensation Additional combined merge candidates Test 3.9a + Test 3.9b + Test 3.9c	Y. Zhang (Qualcomm) H. Huang (Qualcomm)
3.10a 3.10b 3.10c	SATD-based reordering for intra coding SATD-based reordering for inter coding Test 3.10a + Test 3.10b	Y. Wang (Bytedance)

2.2.2. 회의결과 문서 리스트

① **JVET-AI2024** Exploration experiment on enhanced compression beyond VVC capability (EE2)

② **JVET-AI2025** Algorithm description of Enhanced Compression Model 14 (ECM 14)

3. 기존 출원 특허 및 매칭 여부 확인

생략 (공개용 보고서)

4. 향후 회의 일정

회의 차수 (MPEG / JVET)	도시	국가	일정 (MPEG 기준)
148 / 36	Antalya	TR	2024-11-04 to 2024-11-08
149 / 37	Geneva	CH	2025-01-14 to 2025-01-22
150 / 38	Teleconference	-	2025-03-26 to 2025-04-04
151 / 39	Daejeon	KR	2025-06-30 to 2025-07-04
152 / 40	Geneva/CH	CH	2025-09-30 to 2025-10-08

5. 주요 수집자료

생략 (공개용 보고서)

6. 기타

생략 (공개용 보고서)